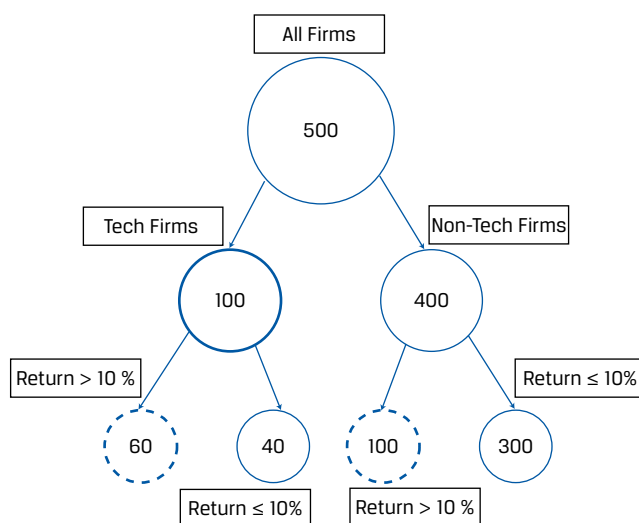


Exhibit 4: Tree Map for Visualizing Bayes' Formula Using Frequencies

$$P(\text{Tech} | R > 10\%) = 60 / (60 + 100)$$

Exhibit 5: Summary of Returns for Tech and Non-Tech Firms in Hypothetical Large-Cap Equity Index

Rate of Return (R)	Type of Firm in Stock Index		Total
	Non-Tech	Tech	
R > 10%	100	60	160
R ≤ 10%	300	40	340
Total	400	100	500

What is the probability a firm is a tech firm given that it has a return of >10 percent or $P(\text{tech} | R > 10\%)$? Looking at the frequencies in the tree map and in the table, we can see many empirical probabilities, such as the following:

- $P(\text{tech}) = 100 / 500 = 0.20$,
- $P(\text{non-tech}) = 400 / 500 = 0.80$,
- $P(R > 10\% | \text{tech}) = 60 / 100 = 0.60$,
- $P(R > 10\% | \text{non-tech}) = 100 / 400 = 0.25$,
- $P(R > 10\%) = 160 / 500 = 0.32$, and, finally,
- $P(\text{tech} | R > 10\%) = 60 / 160 = 0.375$.

This probability is the answer to our initial question.

Without looking at frequencies, let us use Bayes' formula to find the probability that a firm has a return of >10 percent and then the probability that a firm with a return of >10 percent is a tech firm, $P(\text{tech} | R > 10\%)$. First,

$$P(R > 10\%)$$

$$= P(R > 10\% | \text{tech}) \times P(\text{tech}) + P(R > 10\% | \text{non-tech}) \times P(\text{non-tech})$$